

50,000 MIMIC-IV discharge summaries
Gemma-2-2B, layer 16 residual stream, 2,304-d

46 ICD-9 codes per note
assigned by human coders

1 Nine feature sources

SAE dictionaries

ReLU + L1

JumpReLU

GemmaScope (general)

non-SAE, unsupervised

PCA

random directions

TF-IDF n-grams

one feature per ICD code

diff-in-means (LDA)

LR probe

keyword indicators

2 One interface for every source

one [notes $\times$ k] matrix per source; activations max-pooled over tokens

3 One audit for every source

one feature per code selected on shards 0-30, scored on held-out 281-311
point-biserial r , BH-FDR $q = 0.05 \rightarrow$ on-target $|r|$, off-target correlation

4a Concordance

an LLM explains the feature from its top contexts; a blind judge picks the code from 9 same-chapter options

4b Causal ablation

the feature is removed from the residual stream; the change in loss is compared, code-positive vs code-negative notes